

Saturday, May 16, 2026

In the process of porting Emilian Parau's MATLAB code for computing axisymmetric periodic waves on a ferrofluid column to Python. Slowly getting a better understanding of what the code is doing. The mathematics comes from more than one source, but Fu & Il'ichev's "Solitary waves in fluid-filled elastic tubes: existence, persistence, and the role of axial displacement" seems to feature heavily.

My short-term goal is to create a minimal Python script that produces the same output as Parau's MATLAB script. Parau's code utilizes the "Fokas method", which I need to learn (medium-term goal).